BT230001CS

BSH/S-24/CS/AI/DS2T002/01

JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU& RTMNU

Education to Eternity

B.Tech Summer - Summer 2024

(ESE / ~~Make-up~~ / ~~Supplementary~~) Examination

Branch: CSE/AI/CSE-DS Sem:-II

Subject: Advanced Chemistry Subject Code:CS2T002/AI2T002/DS2T002

Marks:100 Date of Exam: 03/06/2024 Time: 3 Hrs.

Instructions:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

COs: Students should be able to:

CO1- Describe various properties of water, describe types of fuel, refining of Petroleum, orbital,electronic configuration, nanomaterial and polymers,energy levels, spectroscopic technique

CO2- Interpret the various classification of fuel, various sources of water, refining of petroleum, classification of CNT, various properties of nanomaterial and polymers, ionization energies

CO3- Apply the Knowledge of characteristics of good fuel, Synthesis of nanomaterial,liquid crystal polymers, zeolite process, Ion exchange process, Hot Lime –Soda process, acid base concept,spectroscopic techniques

CO4- Analyze the question on Proximate and Ultimate analysis of coal, potential use of nanomaterial, phases of thermotropic polymers, analyze question on water characteristics

CO5- Estimate a Modal on commercial grading of coal,synthesis of nanomaterial, advanced polymers, spectroscopic technique, doping

CO6- Organize coal, water as per quality,energy level diagram of diatomic molecules,nanomaterial and polymers.

PARTA

Q. No.	Solve the two words or two sentences (All Questions are compulsory).	Bloom's Level and CO	Marks
Q.1	What is water hardness?	L-1/CO1	2
Q.2	Name two industrial water treatment processes.	L-1/CO1	2
Q.3	List thetypes of boiler troubles.	L-1/CO1	2
Q.4	What is nanoscience?	L-1/CO2	2
Q.5	Name two methods of synthesizing nanomaterials.	L-1/C03	2
Q.6	Define single-walled carbon nanotubes (SWCNTs).	L-1/CO2	2
Q.7	What is the effective nuclear charge?	L-1/CO2	2
Q.8	Define electronic configuration by giving one example.	L-1/CO1	2
Q.9	List the types of orbitals in the periodic table.	L-1/CO2	2

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Q.10	How do s, p, d, and f orbital energies vary across the periodic table?	L-2/CO2	2
Q.11	Show the trend of ionization energy down a group.	L-1/CO2	2
Q.12	What is the principle behind NMR spectroscopy?	L-1/CO1	2

PART B

Q. No.	**Solve the following Descriptive Type Questions (All Questions are compulsory).**	**Bloom's Level and CO**	**Marks**
Q.13	Apply the Calgon process in a domestic water treatment setup.	L-3/CO3	4
Q.14	Identify the cause of sludge formation in a given sample of boiler water.	L-3/CO3	4
Q.15	Select the polarizability of a molecule based on its size and electron configuration.	L-3/CO3	4
Q.16	Make use of molecular orbital theory to draw the molecular orbital diagram for a Hydrogen (H_2) molecule.	L-3/CO6	4
Q.17	Utilize proximate analysis to evaluate a coal sample.	L-3/CO4	4
Q.18	Construct a classification of a series of acids and bases as hard or soft.	L-3/CO3	4

PART C

Q. No.	**Solve the following Descriptive Type Questions (Solve any 04 out of 06).**	**Bloom's Level and CO**	**Marks**
Q.19	Compare the lime-soda process and the ion-exchange process.	L-4/CO3	7
Q.20	Analyze the characteristics of solid, liquid, and gaseous fuels, comparing their properties and differences.	L-4/CO5	7
Q.21	Inspect the trends in atomic size across a period and down a group, comparing and contrasting the observed changes.	L-4/CO3	7
Q.22	Distinguish between the molecular orbital diagrams of H_2 and O_2, comparing their respective features and differences	L-4/CO6	7
Q.23	Compare the effectiveness of NMR spectroscopy with UV spectroscopic techniques.	L-4/CO3	7
Q.24	Determine the best method for preventing caustic embrittlement in a high-pressure boiler.	L-5/CO3	7

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PART D

Q. No.	Solve the following Descriptive Type Questions (Solve any 02 out of 03).	Bloom's Level and CO	Marks
Q.25	Determine the quantities of lime(92 % Pure) and soda (95% Pure) required for the treatment of 120000 liters of hard water containing the following salts-$Ca(HCO_3)_2$= 81 ppm, $CaCl_2$=222ppm, NaCl=5ppm, $MgCl_2$=95ppm, $Mg(HCO_3)_2$=146ppm.	L-5/CO3	12
Q.26	A coal sample was found to contain: C=66.2%, H=4.2%, O=6.1%, N=1.4%, S=2.9%, and remaining is ash. Determine the quantity of dry product of combustion, if 1 kg of coal is burnt with 25% excess air.	L-5/CO4	12
Q.27	Evaluate the impact of nanomaterials on the development of new technologies.	L-5/CO4	12

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